

INTERNSHIP REPORT

DATA ANALYTICS

*A Summer Internship Report submitted in partial fulfillment of the
requirements for the award of degree of*

MASTER OF COMPUTER APPLICATIONS

Submitted

By

TANNETI NAGA SAI

25M11MC149



DEPARTMENT OF COMPUTER APPLICATIONS

ADITYA UNIVERSITY

A.Y. 2026-27

ADITYA UNIVERSITY

DEPARTMENT OF COMPUTER APPLICATIONS



CERTIFICATE

This is to certify that the internship report entitled “ **Data Analytics**” is being submitted by **Tanneti Naga Sai– 25M11MC149** in partial fulfillment of the requirements for the award of the degree of **Master of Computer Applications** in the **Department of Computer Applications** for the academic year **2026–2027**. This report is based on the internship carried out at **HMIES Solutions**, and it reflects the student’s own work, conducted under our guidance and supervision.

Guide/Supervisor Name

Dinesh Kumar H

Mentor

Head of the Department

Dr. Ramesh NS V S C Sripada

Associate Professor & HOD

DECLARATION

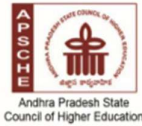
I hereby declare that the internship report entitled “**Predicting Patient Readmission Using Electronic Health Records**” is a genuine and original work submitted to **Aditya University, Surampalem**, in partial fulfillment of the requirements for the award of the **Master of Computer Applications (MCA)** degree.

I further declare that this report has not been submitted, either in full or in part, for the award of any degree or diploma at this or any other educational institution.

Tanneti Naga Sai

25M11MC149

INTERNSHIP CERTIFICATE



HMIES SOLUTIONS PVT. LTD.
(A Division Software Development)



Industrial Internship Certification

This is to certify that Mr/Ms

TANNETI NAGA SAI

Enrolled in the **Department of MCA with Reg. no: 25M11MC149**

From College **ADITYA UNIVERSITY**

Intern ID: **HMIESS2026DA1404**

has Successfully Completed Internship Programme in Physical Mode Titled

“DATA ANALYTICS”

From : 18-05-2026 TO 02-07-2026

Organized by HMIESS in collaboration with
Andhra Pradesh State Council of Higher Education (APSCHE)

DINESH KUMAR HIRAWAT
CEO
HMIES Solution Pvt Ltd



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- **Dr. M Sreenivasa Reddy**, Dy. Pro-Chancellor
- **Dr. M. B. Srinivas**, Vice Chancellor
- **Dr. S Rama Sree**, Pro Vice-Chancellor (Academics)
- **Dr. A Ramesh**, Pro Vice-Chancellor (Engineering & Sciences)
- **Dr. G Suresh**, Registrar
- **Dr. M V Rajesh**, Associate Dean (School of Computing)

Lastly, I would like to thank all the **faculty members, lab technicians, non-teaching staff**, and my **friends** who directly or indirectly helped and supported me in successfully completing my internship project on time.

ABOUT ORGANIZATION

HMIES Solutions Pvt. Ltd. is a software development and IT services company committed to delivering innovative technology solutions and providing industry-oriented internship and training programs for students. The organization focuses on enhancing students' technical knowledge and practical skills by offering hands-on experience in **Data Analytics, Python Programming, Artificial Intelligence (AI), Machine Learning (ML), Data Science, Web Development, and other emerging IT technologies.**

The company believes that practical learning is essential for building successful careers in the technology sector. Through well-structured internship programs, project-based learning, and expert guidance, HMIES Solutions enables students to apply theoretical concepts to real-world business problems. Interns gain exposure to modern software development practices, data analysis techniques, visualization tools, and industry-standard technologies, helping them become confident and competent professionals.

Mission

- To provide high-quality software solutions and reliable IT services.
- To equip students with practical, industry-relevant technical skills through internships and professional training.
- To bridge the gap between academic education and real-world industry requirements.
- To encourage innovation, creativity, and continuous learning among students and professionals.
- To develop skilled, confident, and industry-ready IT professionals through hands-on project experience.

Vision

- To become a leading software development and IT training organization recognized for excellence and innovation.
- To empower students and professionals with advanced technologies and career-focused learning opportunities.
- To contribute to the growth of the IT industry by developing skilled, knowledgeable, and job-ready professionals.
- To build a strong learning ecosystem that promotes technical excellence, research, and innovation.
- To create future technology leaders who can contribute effectively to digital transformation and organizational success.

KEY INITIATIVES

Conduct industry-oriented internship and technical training programs that provide students with practical exposure to real-world IT environments.

Offer hands-on learning opportunities in **Data Analytics, Python Programming, Artificial Intelligence (AI), Machine Learning, and Web Development** through project-based training. Collaborate with **APSCHE** and various educational institutions to enhance students' technical competencies and improve their employability.

Develop and implement real-world projects that strengthen analytical thinking, problem-solving abilities, and software development skills.

Organize workshops, coding sessions, and mentorship programs to keep students updated with the latest industry trends and technologies.

Promote innovation, technical excellence, teamwork, and continuous learning to prepare students for successful careers in the IT sector.

Recognition and Impact

- Recognized for providing quality internship and technical training programs to students.
- Collaborates with APSCHE and educational institutions to support skill development.
- Helps students gain practical knowledge through real-world projects and hands-on learning.
- Enhances students' employability by developing industry-ready technical and professional skills.
- Contributes to building a skilled workforce for the IT and software industry.

Leadership

- HMIES Solutions Pvt. Ltd. is led by experienced professionals with expertise in software development and IT training.
- The leadership team focuses on innovation, quality, and continuous learning.
- They encourage practical, project-based training to prepare students for industry challenges.
- The organization promotes teamwork, professionalism, and ethical work practices.
- Its leaders are committed to developing skilled and career-ready IT professionals.

OVERVIEW

HMIES Solutions Pvt. Ltd. is an IT services and software development company that provides industry-oriented internship and training programs for students. The organization focuses on practical learning in areas such as Data Analytics, Python Programming, Artificial Intelligence, and Web Development. It collaborates with educational institutions and APSCHE to enhance students' technical and professional skills. Through hands-on projects and expert guidance, HMIES prepares students to meet the demands of the modern IT industry.

KEY PROJECT:

Electronic Health Records (EHR) Readmission Analytics Dashboard

The **Electronic Health Records (EHR) Readmission Analytics Dashboard** is a healthcare data analytics project developed using **Python** and **Streamlit**. The primary objective of this project is to analyze patient readmission data and transform raw healthcare records into meaningful insights through interactive dashboards and visualizations. The system helps hospitals and healthcare professionals understand patient readmission patterns, identify high-risk cases, and improve the quality of patient care. The project involves collecting and preprocessing patient data, handling missing values, and performing data analysis using Python libraries such as **Pandas**, **NumPy**, and **Matplotlib**. The processed data is presented through interactive charts, graphs, and Key Performance Indicators (KPIs), making it easier for users to explore healthcare trends and monitor hospital performance.

The dashboard provides valuable insights such as total patient admissions, readmission rates, patient demographics, disease distribution, length of hospital stay, and other healthcare metrics. These visualizations support healthcare administrators and medical professionals in making data-driven decisions, optimizing resource utilization, and improving patient outcomes.

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ABSTRACT

This internship was completed at HMIES Solutions Pvt. Ltd. in the domain of Data Analytics. The main objective of the internship was to gain practical knowledge of data analysis, visualization, and dashboard development using Python. During the internship, I learned to collect, clean, analyze, and visualize data using Python libraries such as Pandas, NumPy, Matplotlib, and Streamlit.

As part of the internship, I developed an Electronic Health Records (EHR) Readmission Analytics Dashboard. The project analyzes patient readmission data and presents meaningful insights through interactive charts, graphs, and Key Performance Indicators (KPIs). The dashboard helps healthcare professionals identify readmission trends and supports data-driven decision-making to improve patient care and hospital performance.

This internship enhanced my technical skills in Python programming, data analytics, data visualization, and problem-solving. It also improved my teamwork, communication, and project management skills while providing valuable exposure to real-world industry practices. Overall, the internship served as an excellent learning experience and strengthened my confidence to pursue a career in the field of data analytics.

WEEK 1 REPORT

1st Week	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	18/05/2026	1	Internship Orientation, Introduction to HMIES Solutions Pvt. Ltd., and Overview of Data Analytics
	19/05/2026	2	Installation and Setup of Python, VS Code, Jupyter Notebook, and Required Libraries
	20/05/2026	3	Python Programming Fundamentals (Variables, Data Types, Operators, Loops, Functions)
	21/05/2026	4	Data Collection, Data Cleaning, and Data Preprocessing using Pandas
	22/05/2026	5	Practiced data type assignments, exploring dynamic typing with primitive types: Integer, Float, String, and Boolean.

Overview:

During the first week of the internship, I was introduced to **HMIES Solutions Pvt. Ltd.** and gained an understanding of the organization's work culture, internship objectives, and the fundamentals of Data Analytics. I installed and configured the required development tools, including Python, VS Code, Jupyter Notebook, and essential Python libraries. I learned the basics of Python programming, such as variables, data types, operators, loops, and functions. I also studied data collection, data cleaning, and preprocessing techniques using the Pandas library. Additionally, I practiced Python data types and explored dynamic typing using primitive data types such as Integer, Float, String, and Boolean.

Conclusion:

The first week provided a strong foundation in Python programming and Data Analytics. I became familiar with the development environment, learned essential programming concepts, and understood the importance of data preprocessing in analytics projects. The practical exercises improved my coding skills and prepared me for advanced topics and project development in the upcoming weeks of the internship.

WEEK 2 REPORT

2nd Week	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	25/05/2026	1	Worked with arithmetic operators (+, -, *, /, //, %, **) and evaluated standard mathematical expressions
	26/05/2026	2	Applied comparison/relational operators and logical operators (and, or, not) to construct compound conditional flags.
	27/05/2026	3	Studied shorthand assignment operators and analyzed operator precedence rules during complex expression parsing.
	28/05/2026	4	Designed linear decision pathways using structured blocks of if, ifelse, and multi-way elif statements.
	29/05/2026	5	Programmed nested decision matrices to solve multi-layered verification challenges, such as determining leap years.

Overview:

During the second week of the internship, I gained a deeper understanding of Python programming concepts by working with different types of operators and decision-making statements. I practiced arithmetic, relational, logical, and assignment operators and learned how operator precedence affects the evaluation of expressions. I also developed programs using conditional statements such as **if**, **if-else**, and **elif** to solve real-world problems. Additionally, I implemented nested conditional statements to perform complex decision-making tasks, including verifying leap years and other logical conditions.

Conclusion:

The second week strengthened my understanding of Python operators and control flow statements. Through hands-on programming exercises, I improved my logical thinking, problem-solving abilities, and coding skills. These concepts provided a strong foundation for writing efficient Python programs and prepared me for more advanced programming and data analytics tasks in the following weeks.

WEEK 3 REPORT

3rd Week	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	1/06/2026	1	Studied the principles of DRY (Don't Repeat Yourself) programming and the importance of modular functions.
	2/06/2026	2	Defined user-specified functions using the <code>def</code> keyword, mapping custom functional scopes and processing executions
	3/06/2026	3	Managed function interaction dynamics by passing positional, keyword, and default arguments, alongside executing explicit return paths
	4/06/2026	4	Analyzed mutable sequences via Lists, executing indexing, slicing, appending, inserting, and element removal operations.
	5/06/2026	5	Explored immutable record sequences using Tuples to understand structural integrity constraints in data pipelines.

Overview:

During the third week of the internship, I learned the importance of writing reusable and organized code using functions in Python. I studied the **DRY (Don't Repeat Yourself)** principle and created user-defined functions using the `def` keyword. I practiced passing positional, keyword, and default arguments to functions and understood how return statements work. In addition, I explored Python data structures such as **Lists** and **Tuples**, performing various operations like indexing, slicing, appending, inserting, updating, and deleting elements. These concepts improved my understanding of data organization and code modularity.

Conclusion:

The third week enhanced my knowledge of Python functions and data structures. I developed the ability to write modular and reusable programs, making the code more efficient and easier to maintain. Working with lists and tuples strengthened my data manipulation skills, which are essential for data analysis and real-world Python applications. These concepts provided a solid foundation for the advanced topics covered in the following weeks.

WEEK-4.REPORT

4th Week	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	8/06/2026	1	Introduction to NumPy and creation of one-dimensional and multi-dimensional arrays
	9/06/2026	2	Array indexing, slicing, reshaping, mathematical operations, and statistical functions
	10/06/2026	3	Introduction to Pandas and creation of Series and DataFrames
	11/06/2026	4	Data cleaning using Pandas: handling missing values, filtering, sorting, and grouping
	12/06/2026	5	Data preprocessing techniques and exploratory data analysis (EDA) using healthcare datasets

Overview:

During the fourth week, I learned the fundamentals of **NumPy** and **Pandas**, which are essential libraries for data analytics. I practiced creating and manipulating arrays, performing mathematical operations, and analyzing datasets. I also explored data cleaning and preprocessing techniques, including handling missing values, filtering records, sorting, grouping, and exploratory data analysis. These activities helped me prepare healthcare data for analysis.

Conclusion:

This week improved my understanding of data manipulation and preprocessing techniques. I gained practical experience in preparing datasets for analysis, which is an important step in building accurate and meaningful analytics applications.

WEEK 5 REPORT

5th Week	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	15/06/2026	1	Introduction to Matplotlib and data visualization concepts
	16/06/2026	2	Creating line charts, bar charts, histograms, and pie charts
	17/06/2026	3	Building interactive dashboards using Streamlit
	18/06/2026	4	Designing Key Performance Indicators (KPIs) and healthcare visualizations
	19/06/2026	5	Integrating charts and analytics into the EHR Dashboard

Overview:

During the fifth week, I focused on **data visualization** and dashboard development. I learned to create various charts using **Matplotlib** and developed interactive web applications using **Streamlit**. I also designed KPI cards and healthcare visualizations to display patient readmission statistics and trends effectively.

Conclusion:

The fifth week enhanced my skills in presenting analytical results through visual dashboards. I learned how to communicate data insights effectively using graphs, KPIs, and interactive user interfaces.

WEEK 6 REPORT

6th Week	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
	22/06/2026	1	Development of the Electronic Health Records (EHR) Readmission Analytics Dashboard
	23/06/2026	2	Testing dashboard features and validating healthcare data
	24/06/2026	3	Performance optimization and debugging of the application
	25/06/2026	4	Final project documentation and dashboard presentation preparation
	26/06/2026	5	Project demonstration, evaluation, and internship completion

Overview:

During the final week, I completed the **Electronic Health Records (EHR) Readmission Analytics Dashboard** by integrating data preprocessing, analysis, visualization, and interactive features. I tested the application, corrected errors, improved its performance, and prepared the project documentation and presentation. Finally, I demonstrated the project and received feedback from the internship mentor.

Conclusion:

The final week marked the successful completion of my internship project. I gained practical experience in Python programming, data analytics, data visualization, and dashboard development. This internship strengthened my technical knowledge, problem-solving abilities, and confidence in applying data analytics techniques to real-world healthcare applications.

OVERALL OUTCOMES

The internship at **HMIES Solutions Pvt. Ltd.** provided me with valuable practical knowledge and hands-on experience in **Python Programming** and **Data Analytics**. Throughout the six-week training program, I strengthened my understanding of Python fundamentals, operators, control statements, functions, data structures, and analytical programming concepts. I also gained practical experience in using **NumPy**, **Pandas**, **Matplotlib**, and **Streamlit** for data analysis, visualization, and dashboard development.

As part of the internship, I successfully developed an **Electronic Health Records (EHR) Readmission Analytics Dashboard**, which analyzes patient readmission data and presents meaningful insights through interactive charts and Key Performance Indicators (KPIs). This project enhanced my ability to collect, clean, process, analyze, and visualize healthcare data while applying real-world problem-solving techniques.

The internship also improved my logical thinking, coding proficiency, communication, teamwork, time management, and documentation skills. Working on industry-oriented tasks helped me understand software development practices and the importance of data-driven decision-making.

Overall, this internship significantly enhanced my technical and professional competencies, increased my confidence in applying data analytics concepts to real-world applications, and prepared me for future opportunities in the fields of **Data Analytics**, **Python Development**, and **Software Engineering**.